

**COURSE 3 - DATA ENGINEERING 2026****C3-064A MeterSpark Protected Review**

Open only after the matching Gate attempt/evidence is frozen.

|                     |                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------|
| <b>Artifact</b>     | C3-064A-REVIEW                                                                                   |
| <b>Status</b>       | 2026 - LEARNER EDITION                                                                           |
| <b>Phase</b>        | C3-05 Distributed Processing with Spark   16 lessons   46-60 focused hours                       |
| <b>Prerequisite</b> | C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here. |
| <b>Current lane</b> | Apache Spark / PySpark 4.2.0   Python 3.10+   Java 17+   learner local[2]                        |
| <b>Brand</b>        | Charcoal #1F2937   Teal #14B8A6   Soft Blue #3B82F6   AMOLED #000                                |

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

## What a passing submission proves

- Correct business grain and source/reject reconciliation before any performance discussion.
- Dimension key is unique before join; no DISTINCT used to hide fan-out.
- Grouped and window outputs have explicit, independently validated grain.
- Plan evidence connects Exchange/join/operator choices to code/data.
- Hot-key diagnosis distinguishes data quality from legitimate skew.
- Performance change has a hypothesis, one controlled variable and correctness parity.
- Tests protect rows/schema/grain/rejection/totals or equivalent case-specific invariants.
- Job is parameterized and reproducible from a clean environment.
- No secrets/private paths/fabricated runtime screenshots are included.

### Routing

**PASS only when every required competency passes and zero critical failures remain. PARTIAL/FAIL -> targeted remediation + changed micro-retest. A fresh full Gate B/C opens only by explicit assignment; failed attempts remain append-only.**